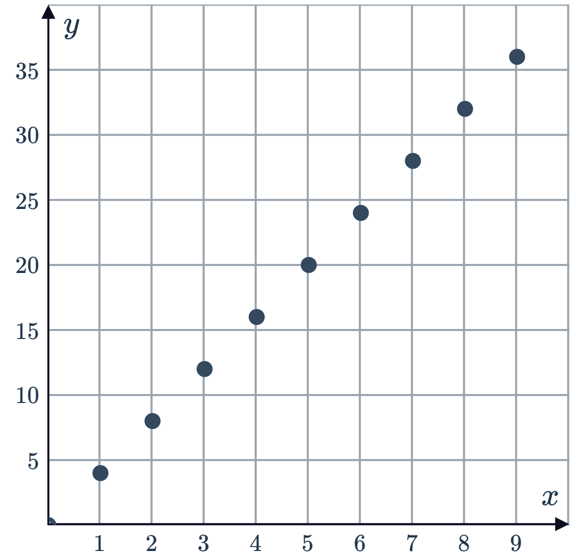
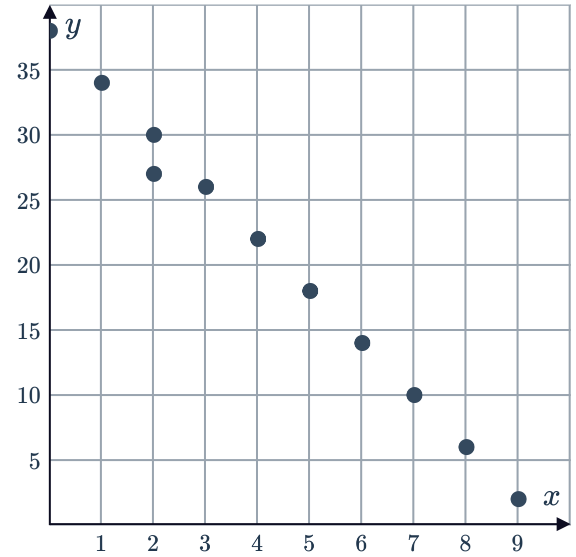
Correlation coefficient

1 For each of the following graphs, write down an estimate of the correlation coefficient:

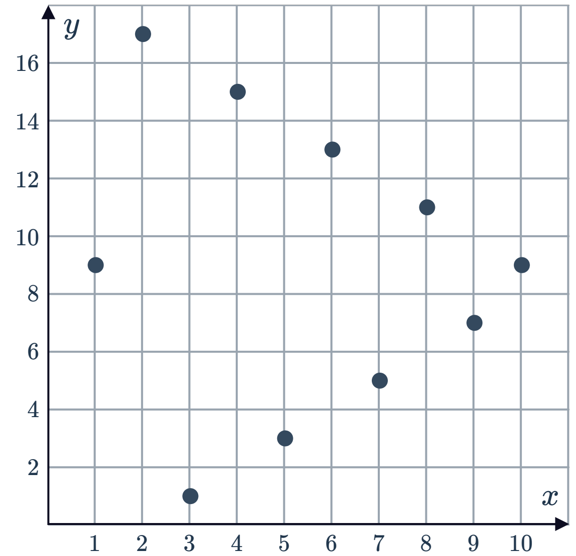
a



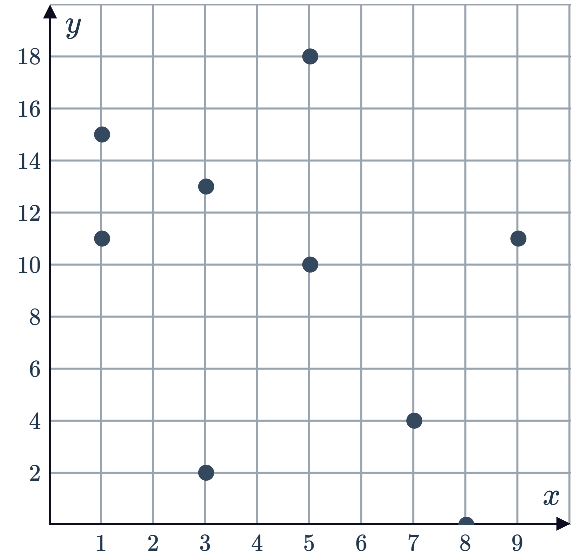
b



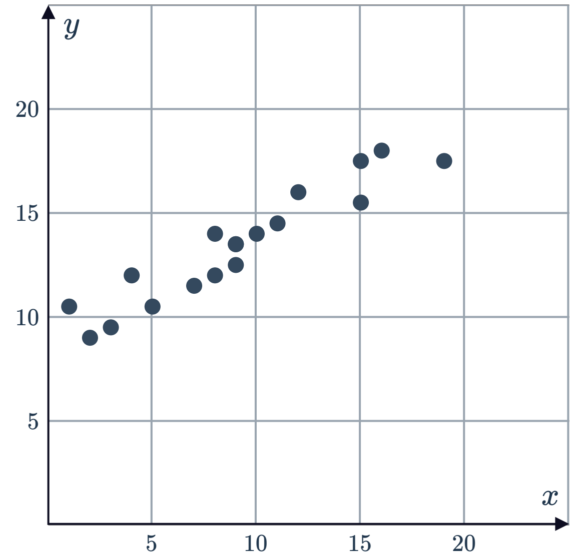
c



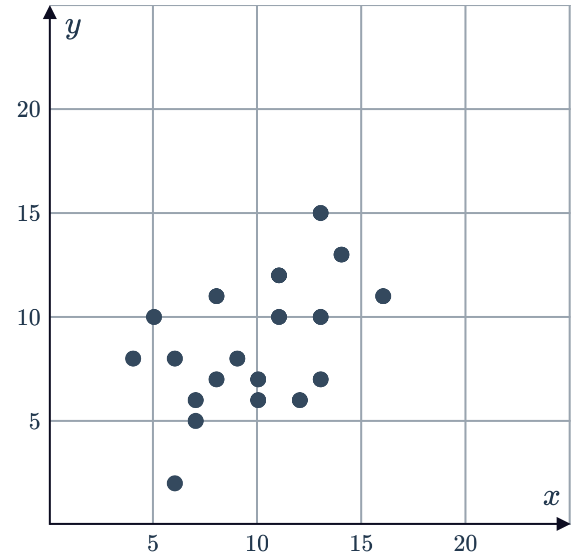
d



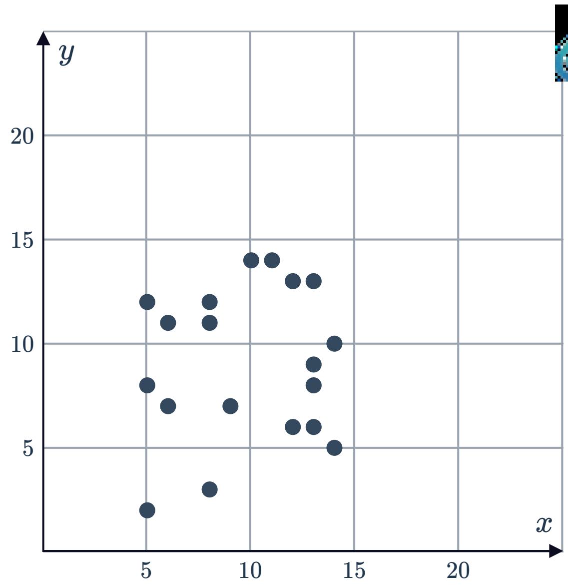
e



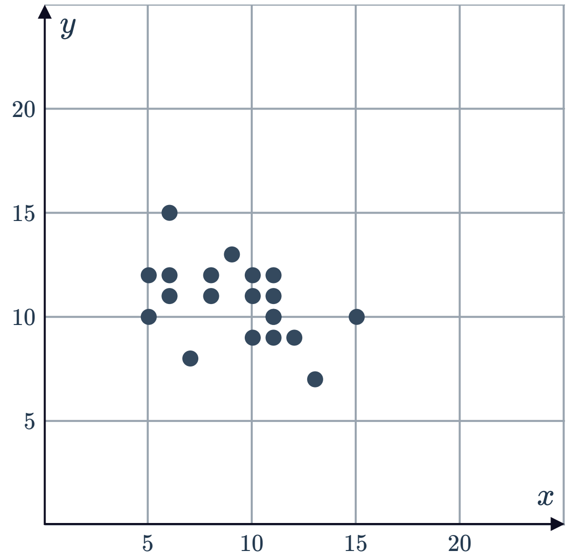
f



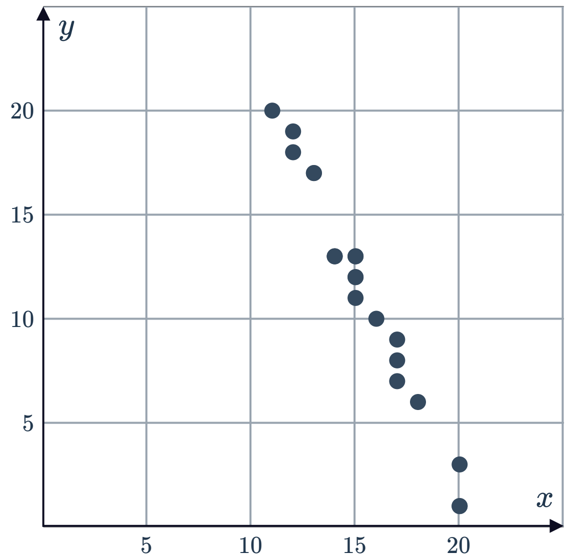
g



h



i



2 If the independent variable increases, describe the effect on the dependent variable for the following studies:

a A study found that the correlation coefficient between heights of women and probability of being turned down for a promotion was found to be -0.90.

b A study found that the correlation coefficient between population of a city and number of speeding fines recorded was found to be 0.83.

c A study found that the correlation coefficient between length of hair and length of fingernails was found to be 0.07.

d A study found that the correlation coefficient between number of bylaws a council has about dog breeding and number of dogs available for adoption at the local shelter was found to be 0.55.

3 For each of the following sets of data:

i Use technology to calculate the correlation coefficient. Round your answer to two decimal places.

ii Describe the correlation between the the two variables.

a

x	3	4	5	6	7	8	9
y	7	7.4	7.88	7.64	7.72	8.2	7.32

b

x	2	6	7	14	17	22
y	2.0	2.4	2.0	0.6	0.9	0.2

c

x	1	4	7	10	13	16	19
y	-1	-1.7	-1.77	-1.84	-1.91	-1.98	-2.05

d

x	4	6	8	13	17	21
y	0.4	0.9	0.6	1.7	2.4	1.7

e

x	4	5	9	13	17	21
y	-0.2	-0.7	-0.4	-1.9	-2.4	-0.9

f

x	2	6	9	12	15	22
y	0	1.5	1.2	2.9	3.7	0.2

g

x	3	6	9	12	15	18	21
y	-7	-7.35	-7.77	-7.56	-7.63	-8.05	-7.28

h

x	9	11	13	15	17	19	21
y	-4	-4.5	-4.55	-4.6	-4.65	-4.7	-4.75

4 Sean is a hotdog vendor. He records the maximum temperature of the day and the number of hotdog sold. The results are in the following table:

Maximum temperature ($^{\circ}$C)	30	34	33	35	33	28	27	31	37	29
Number of hotdogs	18	38	26	40	24	8	20	35	43	38

a Construct a scatter plot to represent the data in the table.

b Calculate the correlation coefficient to two decimal places.

c As the temperature increases, describe what happens to the sales.

Correlation vs causation

5 For each of the following data examples, determine if there is a causal relationship between the variables:

a The number of times a coin lands on heads and the likelihood that it lands on heads on the next flip.

b The amount of weight training a person does and their strength.

6 Determine whether the following describe a causal relationship and not just a correlation:

a An individual's decision to work in construction and his diagnosis of skin cancer.

b The number of minutes spent exercising and the amount of calories burned.

c A decrease in temperature and the increase in attendance at an ice skating rink.

d As a child's weight increases, so does her vocabulary.

7 The table shows the number of fans sold at a store during days of various temperatures:

Temperature ($^{\circ}$C)	6	8	10	12	14	16	18	20
Number of fans sold	12	13	14	17	18	19	21	23

a For this data, will r be greater than zero, less than zero or equal to zero?

b Is there a causal relationship between the variables? Explain your answer.

8 A study found a strong positive association between the temperature and the number of beach drownings.

a Does this mean that the temperature causes people to drown? Explain your answer.

b Is the strong correlation found a coincidence? Explain your answer.

9 A study found a strong correlation betwee [illegible] approximate number of pirates out at sea and the average world temperature.

a Does this mean that the number of pirates out at sea has an impact on world temperature?

b Is the strong correlation found a coincidence? Explain your answer.

c If there is correlation between two variables, is there causation?

10 Irene collects data on the number of baskets of strawberries sold at the roadside each hour one day and the number of cars that drive by in that hour. Her data is shown in the table:
Based on her data, what conclusion can Irene make?

Number of baskets sold	Number of cars
10	31
8	21
2	3
6	21
4	13